CSE472 (Machine Learning Sessional)

Assignment 2: Logistic Regression with Bagging and Stacking

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#1905108

HOW TO RUN:

Ensure the Jupyter Notebook ‘1905108.ipynb’ and Datasets Are in the Same Directory. Install pandas, numpy, scikit-learn, matplotlib, seaborn if not already installed. Then run the notebook ‘1905108.ipynb’.

Dataset 1 is selected as the default. To switch to a different dataset, comment out the current preprocessing function call and uncomment the desired preprocessing function under ‘2.Preprocess dataset’ section. Note that pre1(), pre2(), and pre3() are the preprocessing functions for datasets 1, 2, and 3, respectively.

Preprocess dataset

```
x,y=pre1()
# x,y=pre2()
# x,y=pre3()
# print(x.shape)
x=x.reset_index(drop=True)
y=y.reset_index(drop=True)
```

✓ 0.5s

By default, scaling_type = 'StandardScaler', top_col_no = 20, and MIN_ERR = 0.05 are selected. To change these values, refer to the '1. Preprocessing Functions' section.

```
scaling_type="StandardScaler"
# scaling_type="MinMaxScaler"
top_col_no=20
MIN_ERR=0.05
```

To plot the correlation and scatter plot, uncomment the desired functions under Dataset sections 1(a), 1(b) and 1(c).

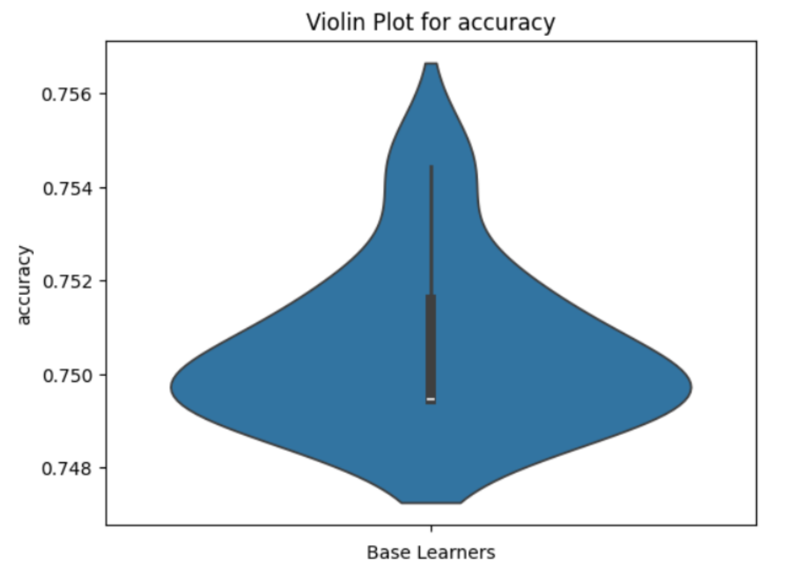
```
# plot_corr(top_correlations)
# print(top_correlations)
top_correlations_features=top_correlations.index
print(top_correlations_features.size)

#scatter plot
# for tf in top_correlations_features:
#     Plot(target_df, features_df, target_col,tf)
```

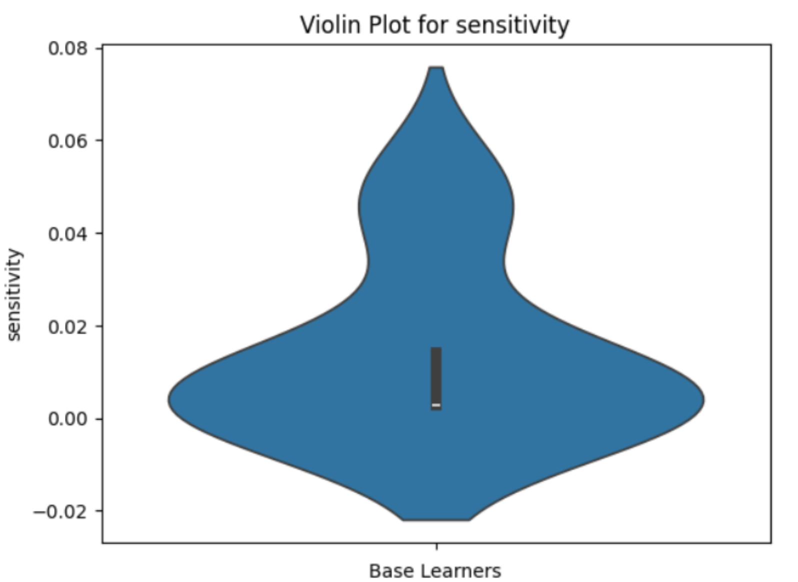
LR, Bagging, Majority voting, Stacking and Performance on test can be found in sections 2, 4, 5, 6 and 7.

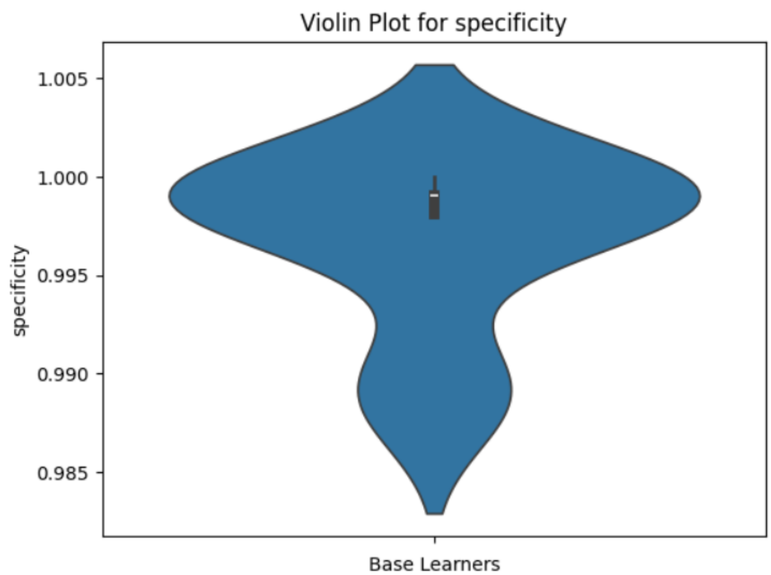
PERFORMANCE EVALUATION:

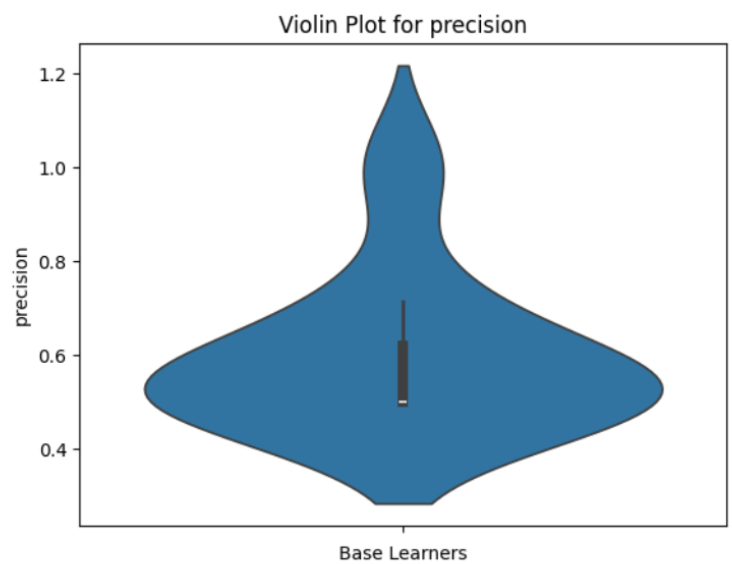
Telco Customer Churn:

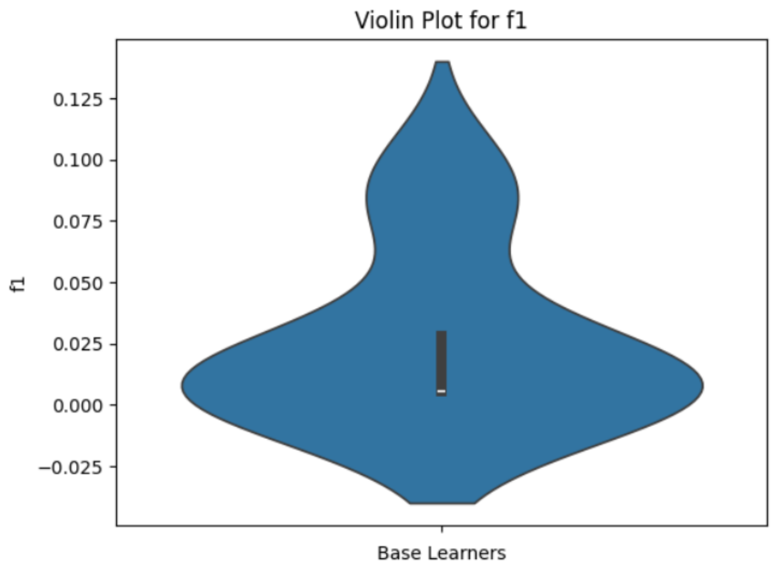
Violin plots:

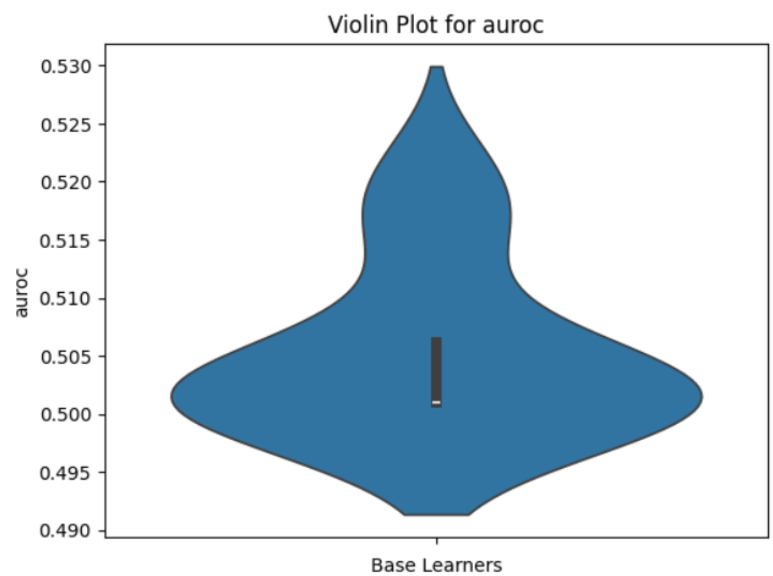


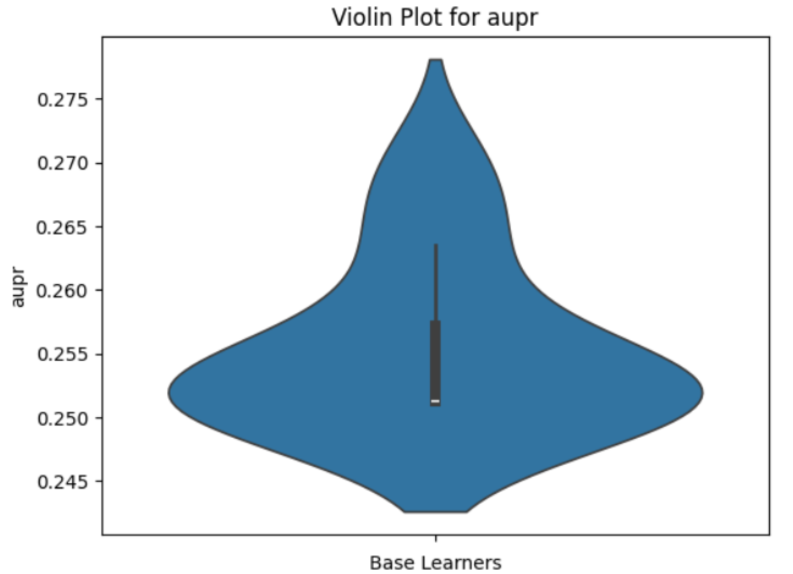



Violin Plot for specificity
1.005
1.000
0.995
0.990
0.985
specificity
Base Learners


Violin Plot for precision
1.2
1.0
0.8
0.6
0.4
precision
Base Learners


Violin Plot for f1
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
f1
Base Learners


Violin Plot for auroc
0.530
0.525
0.520
0.515
0.510
0.505
0.500
0.495
0.490
auroc
Base Learners

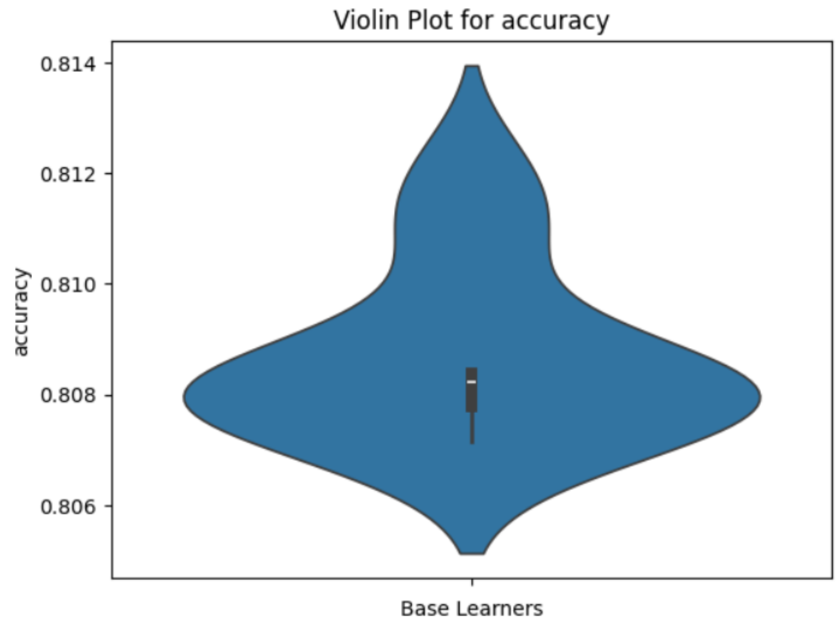


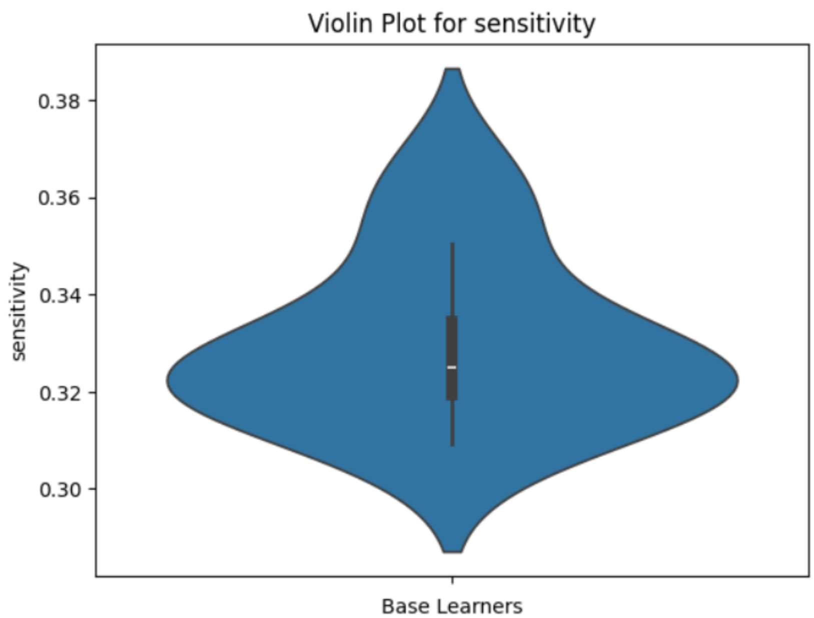
Performance on Test Set

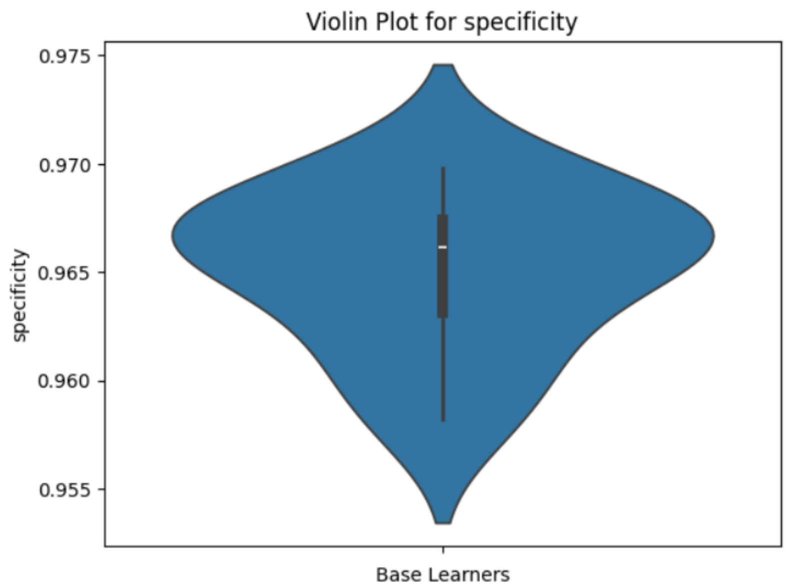
	Accuracy	Sensitivity	Specificity	Precision	F1-score	AUROC	AUPR
LR	0.750573 ± 0.001613	0.013889 ± 0.018089	0.996834 ± 0.004176	0.598948 ± 0.158274	0.026157 ± 0.033339	0.505362 ± 0.007017	0.255443 ± 0.006331
Voting ensemble	0.749466	0.002841	0.999050	0.500000	0.005650	0.500946	0.251243
Stacking ensemble	0.785765	0.269886	0.958215	0.683453	0.386965	0.614050	0.367373

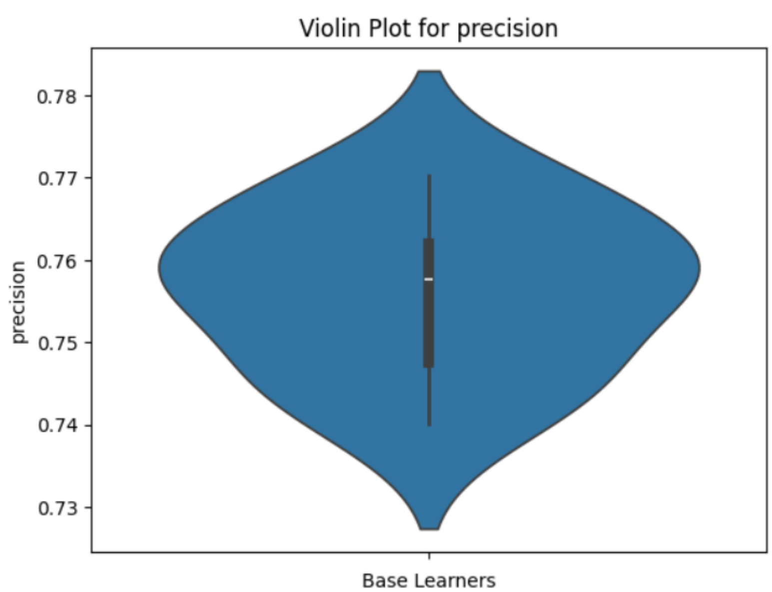
Adult:

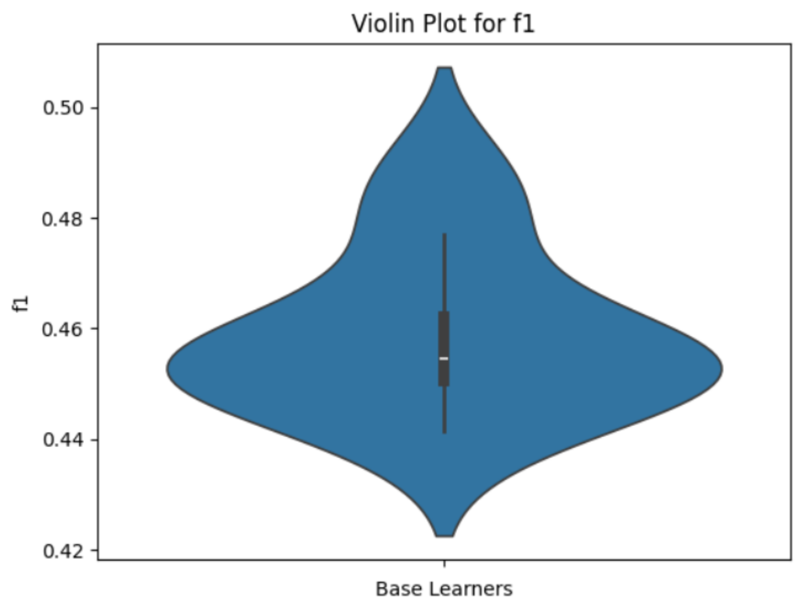
Violin plots:

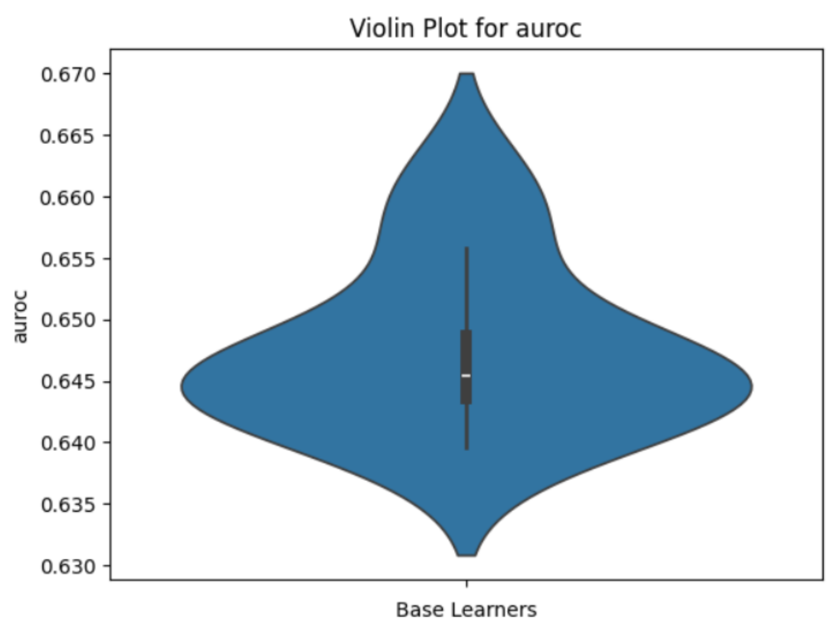


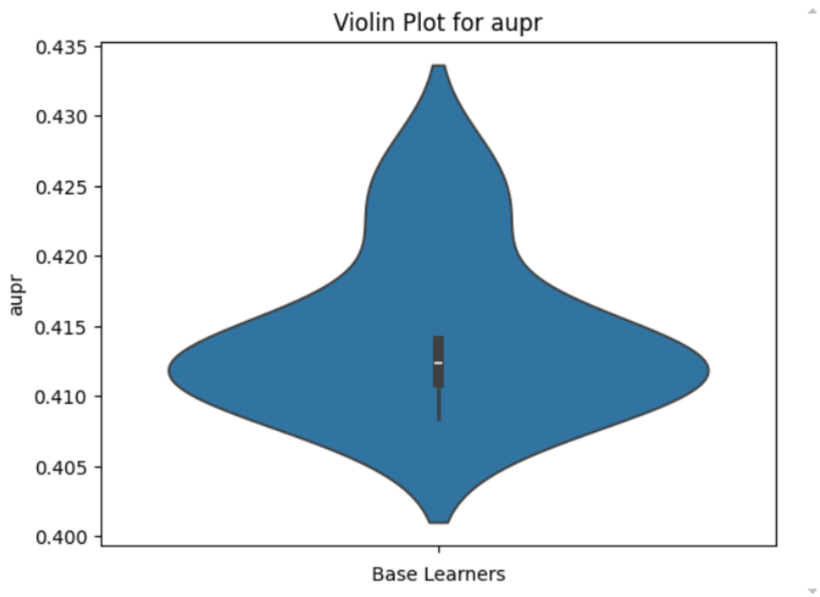



Violin Plot for specificity
0.975
0.970
0.965
0.960
0.955
specificity
Base Learners


Violin Plot for precision
0.78
0.77
0.76
0.75
0.74
0.73
precision
Base Learners


Violin Plot for f1
0.50
0.48
0.46
0.44
0.42
f1
Base Learners


Violin Plot for auroc
0.670
0.665
0.660
0.655
0.650
0.645
0.640
0.635
0.630
auroc
Base Learners

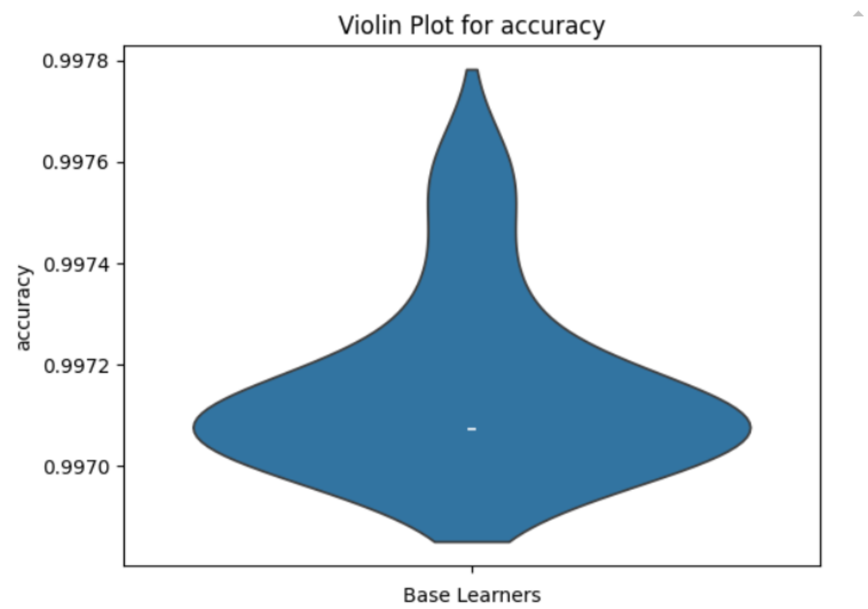


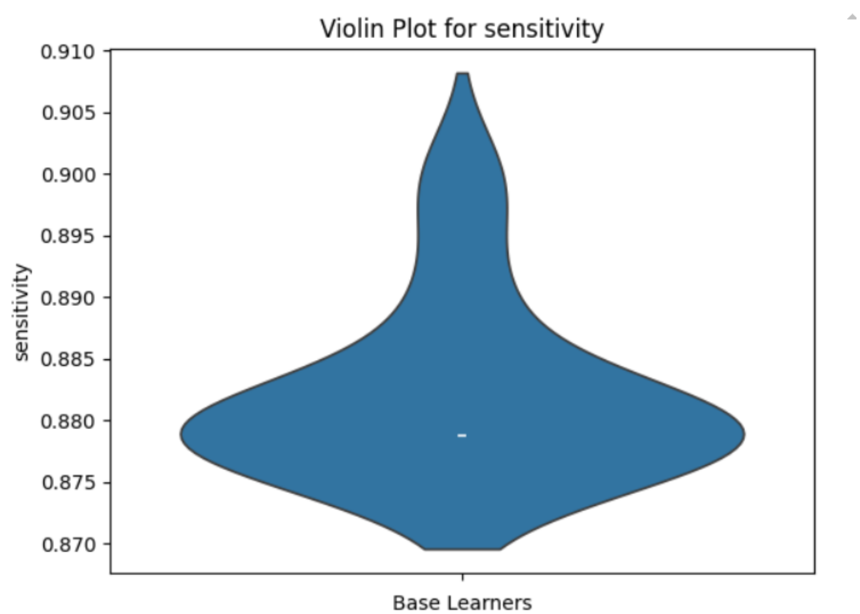
Performance on Test Set

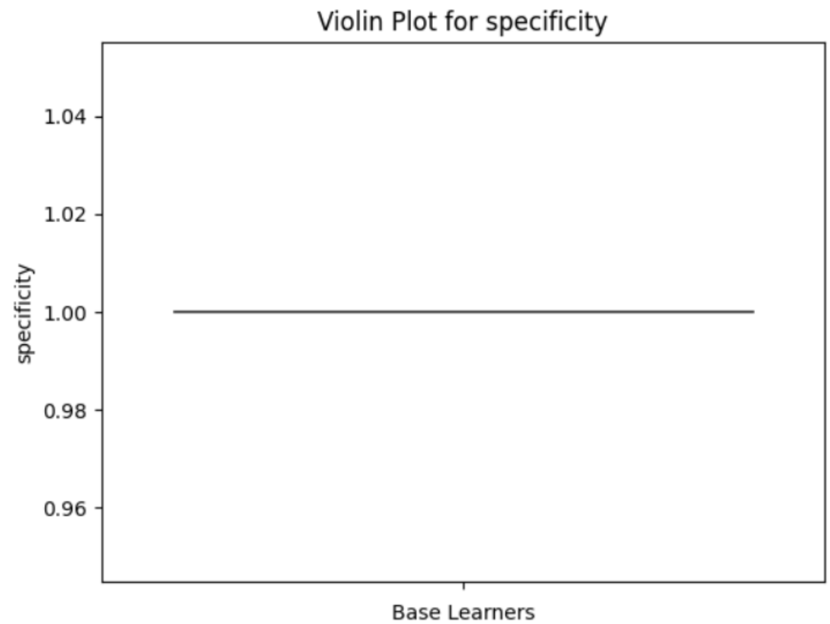
	Accuracy	Sensitivity	Specificity	Precision	F1-score	AUROC	AUPR
LR	0.808680 ± 0.001482	0.329937 ± 0.016303	0.965138 ± 0.003486	0.756227 ± 0.009355	0.459068 ± 0.013886	0.647537 ± 0.006437	0.414409 ± 0.005408
Voting ensemble	0.808543	0.327511	0.965749	0.757576	0.457317	0.646630	0.413757
Stacking ensemble	0.811002	0.352464	0.960856	0.746367	0.478814	0.656660	0.422564

Credit Card:

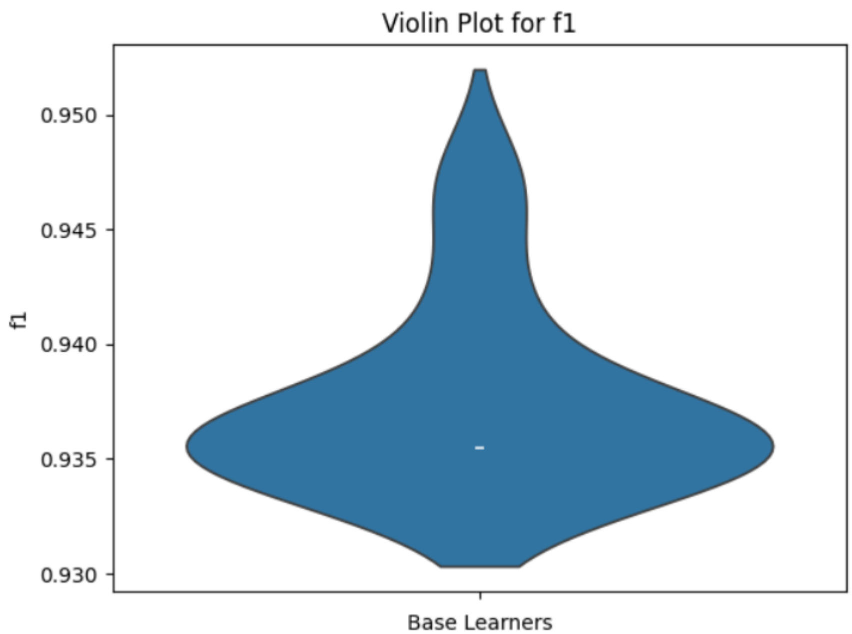
Violin plots:



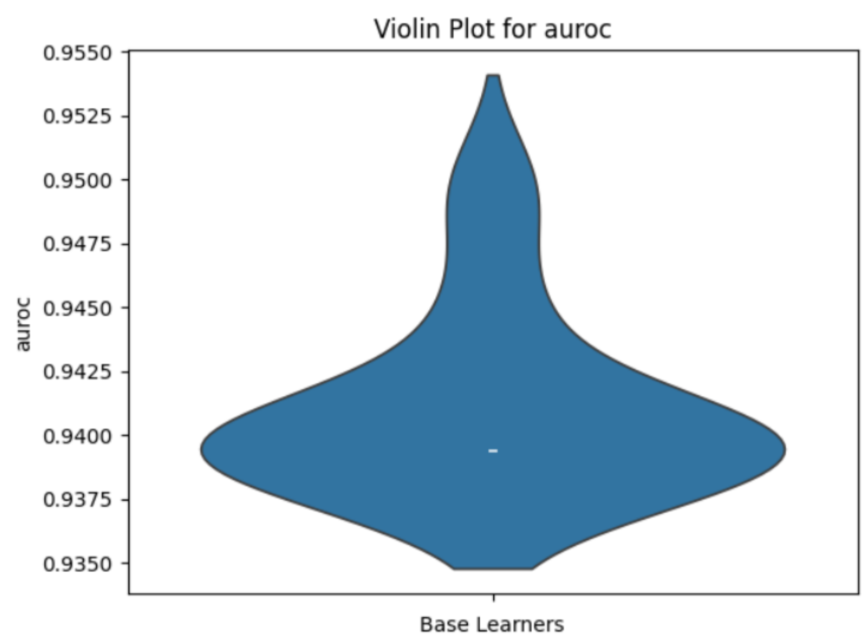



Violin Plot for specificity
1.04
1.02
1.00
0.98
0.96
specificity
Base Learners

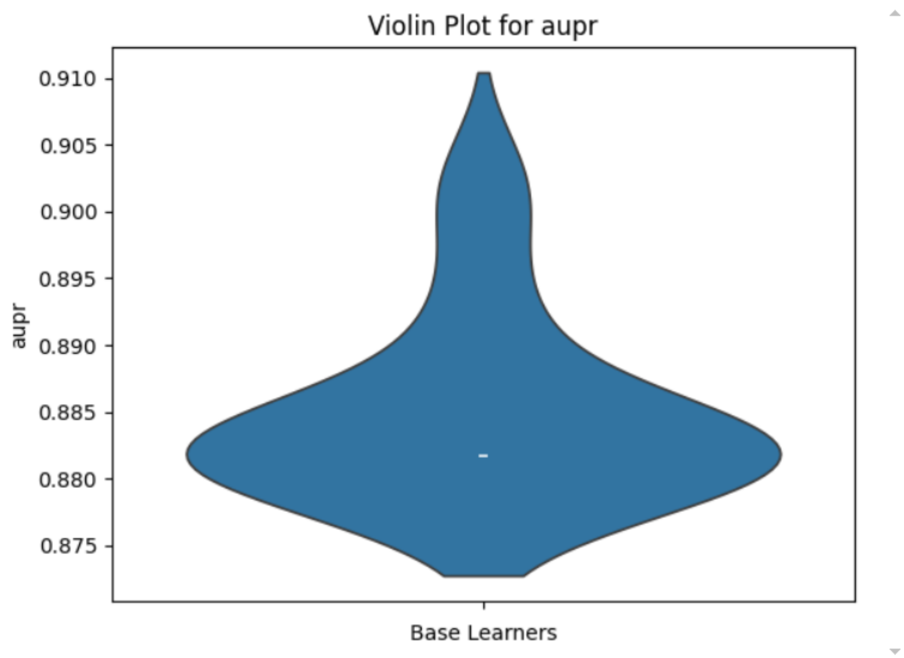

Violin Plot for precision
1.04
1.02
1.00
0.98
0.96
precision
Base Learners



Violin Plot for f1
0.950
0.945
0.940
0.935
0.930
f1
Base Learners



Violin Plot for auroc
0.9550
0.9525
0.9500
0.9475
0.9450
0.9425
0.9400
0.9375
0.9350
auroc
Base Learners



Performance on Test Set

	Accuracy	Sensitivity	Specificity	Precision	F1-score	AUROC	AUPR
LR	0.997154 ± 0.000163	0.882155 ± 0.006734	1.0 ± 0.0	1.0 ± 0.0	0.937375 ± 0.003778	0.941077 ± 0.003367	0.885001 ± 0.006571
Voting ensemble	0.997072	0.878788	1.0	1.0	0.935484	0.939394	0.881715
Stacking ensemble	0.997316	0.888889	1.0	1.0	0.941176	0.944444	0.891572

OBSERVATIONS:

The performance of the base learners varies within a range. May be it is because of sampling. Majority Voting yields results similar to those of the base learners, but the variance among the base learners is mitigated. Overall, Stacking performs better than both the base learners and Majority Voting.